Fossils

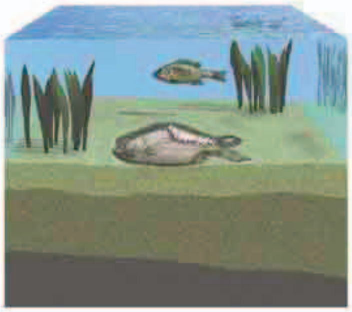
Some sedimentary rocks contain fossils—the remains or traces of once-living things that died long ago. Fossils are typically bones or shells that have turned into stone. Some are beautifully preserved intact skeletons, and some even retain traces of soft tissue. They allow us to reconstruct extinct life-forms such as dinosaurs—if it were not for fossils, we would have no idea that these organisms had ever existed.

FACT!

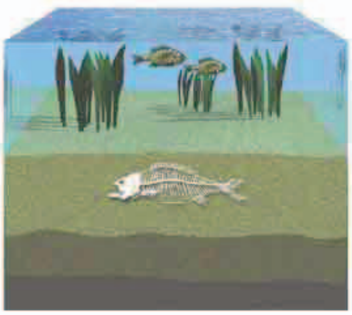
Fossils called coprolites are the remains of animal dung. Scientists have found *Tyrannosaurus rex* coprolites, complete with crunched-up bone fragments from the dinosaur's last meal.

FOSSILIZATION

Most fossils are created by a process that starts when an organism dies and its remains are buried in soft, airless mud. This stops animals from eating it, and prevents rapid decay. Over millions of years, the mud hardens to form sedimentary rock, and minerals dissolved in groundwater slowly turn the organic remains to stone.



◂ DEATH AND BURIAL *A fish living in a lake dies and sinks into the mud on the lake bottom.*



◂ DISSOLVING MINERALS *Over time, more mud settles on top, and minerals start replacing the bones.*

◂ DISCOVERY *The mud becomes rock, which erodes to reveal the fossil.*

TRACE FOSSILS

Some fossils don't preserve the actual remains of the animal or plant. They preserve something else, such as animals' burrows or footprints. These trace fossils can be very revealing. A line of dinosaur footprints, for example, can tell us about the animal's size, its stride length, and how it moved. In other words, they show us how it lived, rather than how it died.